Sangmin Lee

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Research Interest

Massively Multilingual Speech and Natural Language Processing: Automatic speech recognition (ASR), text-to-speech synthesis (TTS), speech translation, and language technologies for low-resource and endangered languages.

Large Speech Language Models (LSLMs): Unified speech–text modeling for multilingual spoken language understanding and generation across diverse languages.

Speech Tokenization and Representation Learning: Learning discrete and continuous speech representations for scalable multilingual and multimodal modeling.

Multilingual Dataset Construction: Building and curating large-scale speech and language resources for low-resource and endangered languages.

Education

Yonsei University — Seoul, South Korea

Ph.D. Program in Electrical and Electronics Engineering (completed as Ph.D. Candidate) — 03/2024 – 02/2027

- Advised by Prof. Hong-Goo Kang, DSP & AI Lab
- Ph.D. candidacy achieved; coursework and qualifying examination completed.
- GPA: 3.94 / 4.3 (equivalent to 4.12 / 4.5)

Korea University — Seoul, South Korea

B.S. in Computer Science and Engineering — 03/2018 – 02/2024

- GPA: 3.76 / 4.5

Publication

Conference Papers

C5. MoLGE: Mixture of Language Group Experts for Efficient Scaling of Massively Multilingual Speech Recognition
Sangmin Lee, Woojin Chung, Woongjib Choi, and Hong-Goo Kang— *COLM 2026* (*Poster*)

C4. UR-BERT: Scaling Text Encoders for Massively Multilingual TTS Through Universal Romanization and Speech Token Prediction
Sangmin Lee, Eekgyun Ahn, Woongjib Choi, and Hong-Goo Kang— *Interspeech 2026* (*Poster*)

C3. UniverSR: Unified and Versatile Audio Super-Resolution via Vocoder-Free Flow Matching
Woongjib Choi, **Sangmin Lee**, Hyeongseob Lim, and Hong-Goo Kang— *ICASSP 2026* (*Poster*)

C2. UniCoM: A Universal Code-Switching Speech Generator
Sangmin Lee, Woojin Chung, Seyun Um, and Hong-Goo Kang— *Findings of EMNLP 2025* (*Poster*)

C1. LAMA-UT: Language Agnostic Multilingual ASR Through Orthography Unification and Language-Specific Transliteration
Sangmin Lee, Woojin Chung, and Hong-Goo Kang— *AAAI 2025* (*Oral*)

Preprints

P4. ELARSpeech: A 30K-Hour Dataset Across 675 Underrepresented Languages for Long-Tail Multilingual Speech Technologies
Sangmin Lee, Yejin Jeon, and David Ifeoluwa Adelani— *arXiv:2610.19388*

P3. AEGIS: Awareness-Enhanced Guidance for Iterative Safeguard
Kyungwon Park, **Sangmin Lee**, Heejae Chon, and Hyungu Kang— *arXiv:2607.17713*

P2. SAGE-LD: Towards Scalable and Generalizable End-to-End Language Diarization via Simulated Data Augmentation
Sangmin Lee, Woongjib Choi, Jihyun Kim, and Hong-Goo Kang— *arXiv:2510.00582*

P1. Talk3D: High-fidelity Talking Portrait Synthesis via Personalized 3D Generative Prior
Jaehoon Ko, Kyusun Cho, Joungbin Lee, Heeji Yoon, **Sangmin Lee**, Sangjun Ahn, and Seungryong Kim— *arXiv:2403.20153*

Awards & Honors

AI Research Computing Support Initiative — Computing Resources
Korea Electronics Technology Institute / Samsung SDS (South Korea) — 06/2026
- Student lead for the project "Efficient Omnilingual Speech Large Language Model (SLLM) via Orthography Unification", awarded access to 16× NVIDIA H100 GPUs (2 nodes).

National Research Foundation Research Grant — Research Funding
National Research Foundation (South Korea) — 03/2026
- Student lead in preparing the proposal for the 3-year project "Efficient Omnilingual Speech Large Language Model (SLLM) via Orthography Unification." (Core Research Type A: $75,000/Year)

NIA Director's Award, 2023 Korean AI Contest Track 1 — Awards
Ministry of Science and ICT (South Korea) — 10/2023
- Collaborated with 4 teammates to train an automatic speech recognition model for dialects, the elderly, and children.

Work Experience

McGill University & Mila — Montréal, Canada
Incoming Research Intern — 12/2026 – 05/2027
- Planning to investigate LSLM for low-resource languages with Prof. David Ifeoluwa Adelani and Dr. Yejin Jeon.

DSP & AI Lab, Yonsei University — Seoul, South Korea
Ph.D. Candidate — 03/2024 – 02/2027
- Researched massively multilingual speech modeling (500+ languages) spanning ASR, TTS, representation learning, and dataset construction.

DSP & AI Lab, Yonsei University — Seoul, South Korea
Undergraduate Research Intern — 01/2024 – 02/2024
- Studied foundational theory and methods in digital signal processing (DSP) and speech signal processing (SSP) in preparation for graduate research.

CV Lab, Korea University — Seoul, South Korea
Undergraduate Research Intern — 08/2023 – 12/2023
- Investigated speech-driven 3D talking-head modeling, focusing on audio-conditioned facial animation.

Republic of Korea Army — Incheon, South Korea
Military Service — 04/2020 – 10/2021
- Completed mandatory military service in the Republic of Korea Army.

Teaching Experience

Department of Electrical and Electronic Engineering, Yonsei University — Seoul, South Korea
Teaching Assistant — 03/2024 – 02/2025
- **EEE2060: Signals and Systems** (Fall 2024)
- **EEE4610: Capstone Design** (Fall 2024)
  - Advised an undergraduate project on deep learning-based emulation of guitar effects.
- **EEE4420: Digital Signal Processing** (Spring 2024)
- **EEE4610: Capstone Design** (Spring 2024)
  - Advised an undergraduate project on machine learning-based personalization of exercise plan recommendations, in collaboration with Hyeonwoo Baek, the founder of PlanFit.

Languages

Korean (Native)
English (Professional working proficiency)
French (Beginner)



Technical Skills

Python
PyTorch
CUDA
Linux
C/C++

